

AquaticVision: Benchmarking Visual SLAM in Underwater Environment with Events and Frames

Yifan Peng*, Yuze Hong*, Ziyang Hong, Apple Pui-Yi Chui, Junfeng Wu

Abstract—Many underwater applications, such as offshore asset inspections, rely on visual inspection and detailed 3D reconstruction. Recent advancements in underwater visual SLAM systems for aquatic environments have garnered significant attention in marine robotics research. However, existing underwater visual SLAM datasets often lack groundtruth trajectory data, making it difficult to objectively compare the performance of different SLAM algorithms based solely on qualitative results or COLMAP reconstruction. In this paper, we present a novel underwater dataset that includes ground truth trajectory data obtained using a motion capture system. Additionally, for the first time, we release visual data that includes both events and frames for benchmarking underwater visual positioning. By providing event camera data, we aim to facilitate the development of more robust and advanced underwater visual SLAM algorithms. The use of event cameras can help mitigate challenges posed by extremely low light or hazy underwater conditions. The webpage of our dataset is <https://sites.google.com/view/aquaticvision-lias>.

I. INTRODUCTION

With the deepening exploration of oceans, underwater robotics applications have gained significant prominence. Visual SLAM technology provides robots with reliable localization and environmental perception capabilities in GPS-denied underwater environments at relatively low cost, serving as a crucial foundation for intelligent autonomous underwater robotics. Unlike terrestrial environments, visual SLAM in underwater scenarios faces substantially more challenges, including unpredictable lighting conditions, variable water clarity, and abundant unstructured textures. Consequently, conventional visual cameras alone cannot adequately address underwater challenges, prompting researchers to explore novel sensor technologies to enhance the robustness and accuracy of underwater visual SLAM systems.

In recent years, event camera has attracted considerable academic attention as bio-inspired visual sensors due to their high dynamic range (HDR), low latency, and high temporal resolution. Unlike traditional frame-based cameras that capture entire scenes at fixed frame rates, the event camera asynchronously records intensity changes at each pixel, generating data only when brightness variations exceed predetermined thresholds. This operating mechanism renders

event camera data more robust in underwater environments, opening new possibilities for underwater visual SLAM to some extent. The complementary strengths of event camera and traditional camera may offer a meaningful pathway for developing more robust underwater visual SLAM systems.

Most existing underwater datasets focus on deep learning tasks such as semantic segmentation and object detection, as in [1] [2]. Additionally, [3] provides a dataset for underwater color correction and depth estimation. SQUID [4] is a dataset for 3D scene reconstruction. Currently, there are few datasets available for validating underwater visual SLAM, with the dataset released in [5] usable for VIO system validation, but without providing groundtruth. Recently, the FLSea dataset proposed in [6] contains stereo images and IMU data collected in real underwater environments and provides groundtruth, making it suitable for validating both SLAM and depth estimation algorithms. However, as mentioned in [6], the pose groundtruth in FLSea is not captured using motion capture systems and may contain slight imperfections.

To provide more reliable groundtruth for underwater visual SLAM algorithms and to facilitate further advancements in this field, **this paper introduces the first underwater dataset that simultaneously incorporates event camera data, traditional camera frames, and IMU measurements. Furthermore, we release groundtruth trajectories tracked by a motion capture system**, enabling researchers to validate the localization accuracy and performance of their algorithms in underwater environments.

II. OVERVIEW

A. Sensors setup

The DAVIS346 Stereo Kit serves as our comprehensive data acquisition system. This sensor integrates dual DAVIS346-Mono event cameras capable of simultaneously capturing event streams and grayscale images. MPU-6500 IMU is embedded within the sensor, providing high-precision motion sensing capabilities with 3-axis gyroscope and 3-axis accelerometer measurements.

The resulting dataset features high-temporal-resolution stereo event data sampled at 1000 Hz, complemented by stereo grayscale frames captured at 20 Hz with a spatial resolution of 346×260 pixels. Additionally, the system incorporates IMU data synchronized at 200 Hz. Timestamps between all sensors are synchronized in hardware. Besides, for all data sequences published in our dataset, the groundtruth trajectories are acquired by using CHINGMU-U4 motion capture system. It provides 6D pose measurements of a specific coordinates frame, which is defined by a set

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AquaticVision: 使用事件和帧在水下环境中对视觉SLAM进行基准测试

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Abstract—许多水下应用, 例如海上资产检查, 依赖于视觉检查和详细的三维重建。最近在水生环境下的水下视觉SLAM系统方面取得的进展, 在海洋机器人研究中引起了重大关注。然而, 现有的水下视觉SLAM数据集通常缺乏真实轨迹数据, 这使得仅基于定性结果或COLMAP重建来客观比较不同SLAM算法的性能变得困难。在本文中, 我们提出了一个新颖的水下数据集, 其中包括使用动作捕捉系统获得的真实轨迹数据。此外, 我们首次发布了包含事件和帧的视觉数据, 用于水下视觉定位的基准测试。通过提供事件相机数据, 我们旨在促进更健壮和先进的水下视觉SLAM算法的发展。使用事件相机可以帮助缓解由极低光照条件带来的挑战

事件相机数据在水下环境中更为稳健, 在一定程度上为水下视觉SLAM开辟了新的可能性。事件相机和传统相机的互补优势可能为开发更稳健的水下视觉SLAM系统提供有意义的途径。

大多数现有的水下数据集侧重于深度学习任务, 如语义分割和目标检测, 如文献 [1][2] 所示。此外, [3] 提供了用于水下颜色校正和深度估计的数据集。SQUID [4] 是用于三维场景重建的数据集。目前, 用于验证水下视觉 SLAM 的数据集很少, 其中 [5] 发布的数据集可用于 VIO 系统验证, 但未提供真实值。最近, 文献 [6] 提出的 FLSea 数据集包含在真实水下环境中采集的立体图像和 IMU 数据, 并提供真实值, 使其适用于验证 SLAM 和深度估计算法。然而, 如 [6] 所述, FLSea 中的位姿真实值并非通过运动捕捉系统采集, 可能存在轻微误差。

一、引言

随着对海洋的深入探索, 水下机器人应用得到了显著发展。视觉SLAM技术为机器人在无法使用GPS的水下环境中提供了可靠的定位和环境感知能力, 成本相对较低, 是智能自主水下机器人发展的关键基础。与陆地环境不同, 水下场景中的视觉SLAM面临更多挑战, 包括不可预测的光照条件、多变的水体清澈度以及丰富的非结构化纹理。因此, 仅依靠传统的视觉相机无法充分应对水下挑战, 这促使研究人员探索新型传感技术, 以提高水下视觉SLAM系统的鲁棒性和精度。

近年来, 事件相机作为受生物启发的视觉传感器, 因其高动态范围 (HDR)、低延迟和高时间分辨率而吸引了大量学术关注。与以固定帧率捕捉整个场景的传统帧式相机不同, 事件相机在每个像素上异步记录强度变化, 仅在亮度变化超过预定阈值时生成数据。这种操作机制使得

为了为水下视觉 SLAM 算法提供更可靠的真实数据, 并促进该领域的进一步发展, 本文引入了首个同时包含事件相机数据、传统相机帧和 IMU 测量的水下数据集。此外, 我们发布了由动作捕捉系统跟踪的真实轨迹, 使研究人员能够验证其算法在水下环境中的定位精度和性能。

二、概述

A. Sensors setup

DAVIS346 立体套件作为我们综合的数据采集系统。该传感器集成了双 DAVIS346-Mono 事件相机, 能够同时捕捉事件流和灰度图像。传感器内置 MPU-6500 IMU, 提供高精度的运动感测能力, 包括三轴陀螺仪和三轴加速度计测量。

生成的数据集具有高时间分辨率的立体事件数据, 采样率为1000 Hz, 并辅以以20 Hz采集的立体灰度帧, 空间分辨率为 346×260 像素。此外, 系统还集成了以200 Hz同步的IMU数据。所有传感器之间的时间戳在硬件上实现同步。此外, 对于我们数据集中发布的所有数据序列, 真实轨迹是使用CHINGMU-U4动作捕捉系统获取的。它提供特定坐标系的6自由度位姿测量, 该坐标系由一组定义

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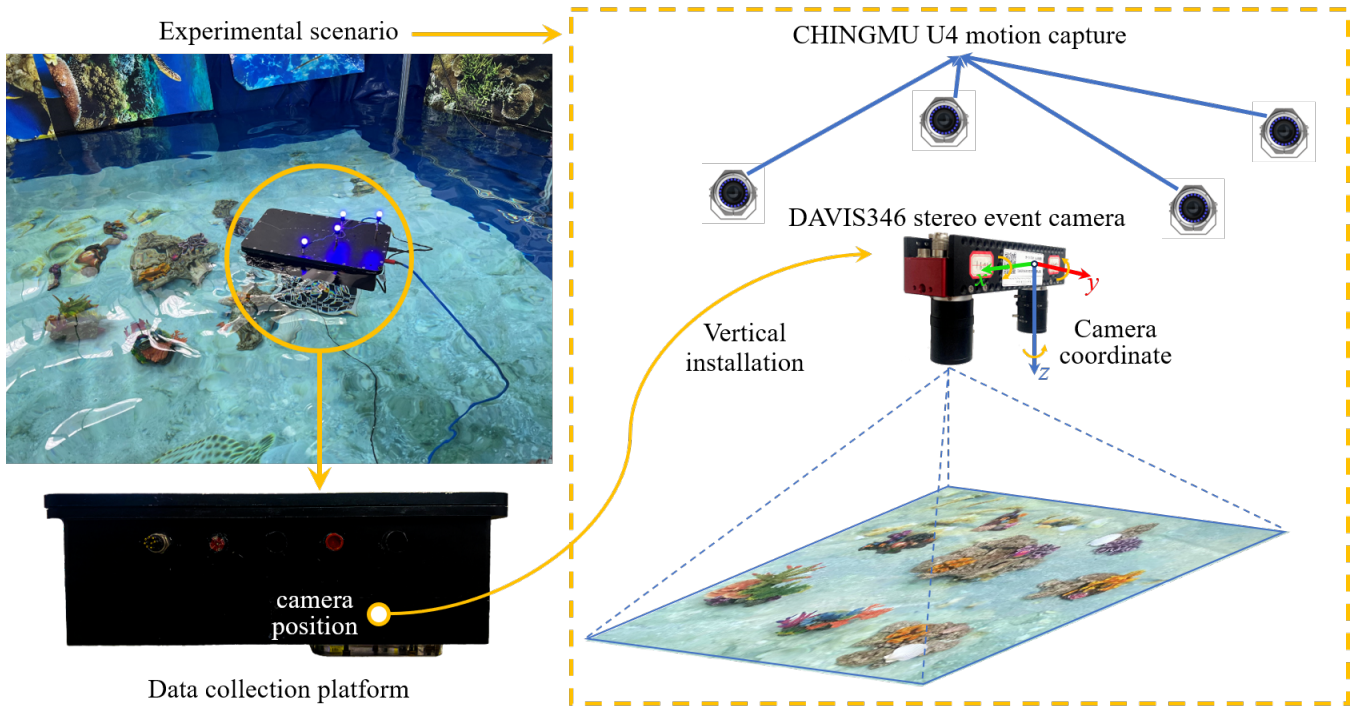


Fig. 1: Experimental scenario and underwater data collection platform. IMU coordinates align with camera one.

of marker attached on the top of data collection platform. The pose measurements are recorded at a rate of 120 Hz. Table I summarizes all sensors used and their data publishing frequency.

TABLE I: Sensors used in the experiments

Sensor	Type	Frequency
Event camera	DAVIS346	1000Hz
Frame camera	DAVIS346	20Hz/30Hz
IMU	MPU-6500	200Hz/1000Hz
Motion capture	CHINGMU-U4	120Hz

B. Experimental Scenario

The experimental scene is a pool with dimensions of $4 \times 3 \times 2$ meters, with a water depth of approximately 1 meter. The pool bottom is covered with a real seabed beach image as background, and multiple coral models are placed to simulate the underwater environment to the greatest extent possible. Vertically installed in the data collection platform with its z-axis oriented toward the pool bottom, the DAVIS346 event camera captures data as the platform navigates within a 3×2 meter planar area. Four motion capture cameras are set up above the pool to obtain groundtruth. Fig. 1 illustrates the experimental scenario and the data collection platform. It should be noted that the motion capture equipments are not installed underwater, which is done to expand the robot's range of motion. The collection platform is actually floating on the water surface, with markers installed above it, higher than the water level.

Based on the visual feature characteristics of the scenes, we divide the collected 9 data sequences into two categories:

easy and hard. Sequences 01 through 05 belong to the easy category, featuring clear visual characteristics that simulate underwater environments with good lighting and water quality. Sequences 06 through 09 belong to the hard category, including three unique underwater environmental challenges: darkness, HDR (High Dynamic Range), and blur. TABLE II shows the characteristics of all nine data sequences. The with_board or no_board suffix indicates whether a calibration board is present underwater.

C. Calibration

The DAVIS integrates both conventional imaging and event sensing capabilities on a unified pixel array, allowing for a streamlined calibration workflow. This architectural advantage enables researchers to employ established calibration frameworks like Kalibr [7] on the standard image output, with the resulting calibration parameters being directly transferable to the event component without requiring separate calibration procedures. The extrinsic parameters between the camera and IMU were calibrated by using Kalibr [7].

D. Data Format

The proposed dataset is organized as follow:

```
dataset_name
├── calibration
│   ├── cam0_pinhole.yaml
│   ├── cam1_pinhole.yaml
│   └── davis_imucam_underwater.yaml
└── data_sequences
    ├── 01_Scan_with_board
    └── ...
```

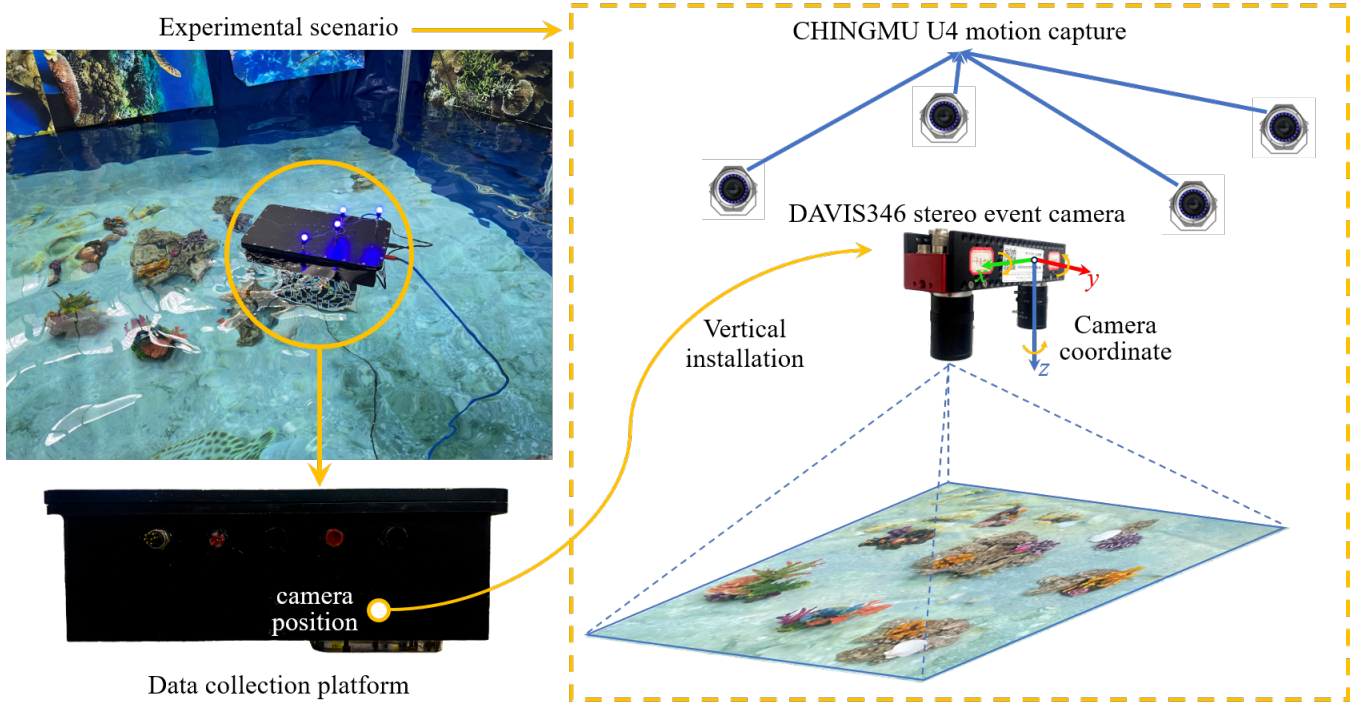


图 1: 实验场景 里约和 水下数据采集平台。IMU坐标 与相机一对齐。

标记附着在数据采集平台的顶部。位姿测量以 120 Hz 的频率记录。表 I 总结了所有使用的传感器及其数据发布频率。

表 I: 实验中使用的传感器

Sensor	Type	Frequency
Event camera	DAVIS346	1000Hz
Frame camera	DAVIS346	20Hz/30Hz
IMU	MPU-6500	200Hz/1000Hz
Motion capture	CHINGMU-U4	120Hz

B. Experimental Scenario

实验场景是一个尺寸为 $4 \times 3 \times 2$ 米的水池，水深约为 1 米。池底覆盖有真实海底沙滩的图像作为背景，并放置多个珊瑚模型，以尽可能模拟水下环境。DAVIS346 事件相机垂直安装在数据采集平台上，其 z 轴朝向池底，在平台在 3×2 平面区域内运动时捕捉数据。池上方设置了四台动作捕捉相机以获取真实值。图 1 展示了实验场景和数据采集平台。需要注意的是，动作捕捉设备没有安装在水下，这是为了扩大机器人运动范围。采集平台实际上浮在水面上，标记安装在平台上方，高于水面水平。

根据场景的视觉特征，我们将收集的9个数据序列分为两类：

简单和困难。序列01到05属于简单类别，具有清晰的视觉特征，模拟水下环境，并具有良好的照明和水质。序列06到09属于困难类别，包括三种独特的水下环境挑战：黑暗、高动态范围(HDR)和模糊。表II显示了所有九个数据序列的特征。带板或无板的后缀表示水下是否存在校准板。

C. Calibration

DAVIS 在统一的像素阵列上集成了传统成像和事件感知能力，从而实现了简化的校准工作流程。这一架构优势使研究人员能够在标准图像输出上使用像 Kalibr [7] 这样的既有校准框架，生成的校准参数可以直接转移到事件组件，无需单独的校准过程。相机与 IMU 之间的外参通过使用 Kalibr [7] 进行了校准。

D. Data Format

The proposed dataset is orgnized as follow:

```

dataset_name
├── calibration
│   ├── cam0_pinhole.yaml
│   ├── cam1_pinhole.yaml
│   └── davis_imucam_underwater.yaml
└── data_sequences
    ├── 01_Scan_with_board
    └── ...
  
```

TABLE II: Data sequences characteristics

Sequence	Name	Category	Duration	Description
01	Scan_with.board	Easy	93.32s	Clear water, corals evenly distributed, trajectory covers underwater area
02	Cross1_with.board	Easy	109.00s	Clear water, corals centrally distributed, cross-pattern trajectory
03	Cross2_no.board	Easy	68.03s	Clear water, corals evenly distributed, cross-pattern trajectory
04	Loop1_with.board	Easy	75.70s	Clear water, corals evenly distributed, loop-shaped trajectory
05	Loop2_no.board	Easy	59.92s	Clear water, corals centrally distributed, loop-shaped trajectory
06	Dark1_with.board	Hard	75.04s	Clear water, low-light condition
07	Dark2_with.board	Hard	74.30s	Clear water, low-light condition
08	HDR	Hard	97.00s	High dynamic range lighting conditions creating challenging exposure variations
09	Blur	Hard	133.47s	Water turbidity affecting image clarity and feature detection

TABLE III: ATE OF SLAM SYSTEMS ON SAMPLE SEQUENCES [m]

Method	S01	S02	S03	S04	S05	S06	S07	S08	S09
VINS-Stereo	0.0573	0.0616	0.0428	0.0489	0.0907	1.1629	0.0698	failed	0.0507
ORB-SLAM2	0.4231	0.1134	0.0864	0.4267	0.1712	failed	failed	0.1763	0.1546
ESVO2	failed	failed	failed	failed	failed	failed	failed	failed	failed

└─ 09_Blur

In the calibration folder, there are three calibration files in YAML format, which contain the intrinsic parameters of the left camera (cam0) and right camera (cam1), as well as the extrinsic parameters between the cameras and the IMU. The data_sequences folder contains data for 9 sequences.

Each sequence provides sensor data, groundtruth, and grayscale images. Event, image, and IMU data are provided in the rosbag format, including two types of rosbags: one with images at 30Hz and IMU at 1000Hz, and another with images at 20Hz and IMU at 200Hz. Users can choose the appropriate rosbag for their algorithms and perform down-sampling operations when needed. The baseline trajectory is provided in the CSV file. Additionally, we have released a groundtruth file in TUM format to facilitate evaluation based on the evo tool. Furthermore, the left and right camera grayscale images at 30Hz frame rate have been extracted and stored separately in the l1 and r1 folders, accompanied by a timestamp alignment file. The data sequence is organized as follow:

```

id.sequence.name
├── data rosbag
│   ├── name_imu_1000hz_images_30hz.bag
│   └── name_imu_200hz_images_20hz.bag
├── groundtruth
│   ├── gt.csv
│   └── gt.tum
└── Stereo images
    ├── l1
    ├── r1
    └── timestamp_pairs.txt

```

E. Baseline trajectory using visual SLAM

Various mainstream SLAM systems are tested across nine sequences from the proposed dataset, with detailed characteristics presented in Table II. To assess performance, we employ the Absolute Trajectory Error (ATE) as our primary metric. For consistent comparison, we utilize the evo tool to

align each estimated trajectory with its corresponding ground truth, thereby calculating precise ATE measurements. The quantitative results are shown in Table III.

To evaluate the performance of frame-based visual SLAM in underwater environments, Vins-Stereo-fusion [8] and ORB-SLAM2 [9] are used for testing. The testing of Vins is conducted using rosbags with IMU data at 200Hz and images at 20Hz, while the testing of ORB-SLAM2 utilizes grayscale images at 30Hz. It should be specifically noted that the purpose of selecting ORB-SLAM2 for testing is to assess the performance difference between visual odometry (VO) and visual-inertial odometry (VIO) on the proposed underwater dataset. The state-of-the-art event-based SLAM algorithm, ESVO2 [10], is also tested to evaluate the performance of event camera in challenging underwater environments.

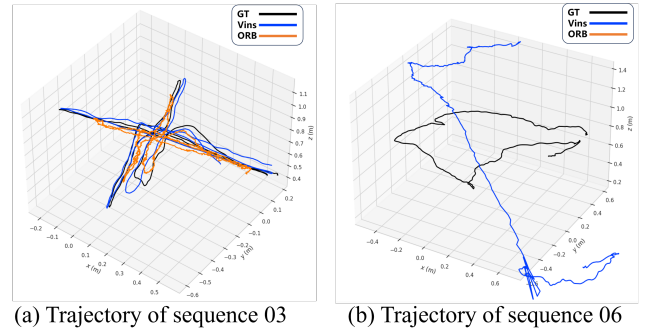


Fig. 2: Estimated and groundtruth (GT) trajectories of 2 sample sequences.

1) Frame-based visual SLAM:

As shown in Table III, test results based on the easy class sequences (S01-S05) indicate that the pose estimation accuracy of the VIO system (VINS-Stereo) is significantly higher than that of the VO system (ORB-SLAM2). This is because the presence of water waves and buoyancy causes high-frequency rapid jitter of the data collection platform during turns. On one hand, this violates the constant velocity

表 II: 数据序列特征

Sequence	Name	Category	Duration	Description
01	Scan_with_board	Easy	93.32s	Clear water, corals evenly distributed, trajectory covers underwater area
02	Cross1_with_board	Easy	109.00s	Clear water, corals centrally distributed, cross-pattern trajectory
03	Cross2_no_board	Easy	68.03s	Clear water, corals evenly distributed, cross-pattern trajectory
04	Loop1_with_board	Easy	75.70s	Clear water, corals evenly distributed, loop-shaped trajectory
05	Loop2_no_board	Easy	59.92s	Clear water, corals centrally distributed, loop-shaped trajectory
06	Dark1_with_board	Hard	75.04s	Clear water, low-light condition
07	Dark2_with_board	Hard	74.30s	Clear water, low-light condition
08	HDR	Hard	97.00s	High dynamic range lighting conditions creating challenging exposure variations
09	Blur	Hard	133.47s	Water turbidity affecting image clarity and feature detection

表 III: 样本序列上 SLAM 系统的 ATE [米]

Method	S01	S02	S03	S04	S05	S06	S07	S08	S09
VINS-Stereo	0.0573	0.0616	0.0428	0.0489	0.0907	1.1629	0.0698	failed	0.0507
ORB-SLAM2	0.4231	0.1134	0.0864	0.4267	0.1712	failed	failed	0.1763	0.1546
ESVO2	failed	failed	failed	failed	failed	failed	failed	failed	failed

09 模糊

在校准文件夹中, 有三个 YAML 格式的校准文件, 其中包含左摄像头 (cam0) 和右摄像头 (cam1) 的内参, 以及摄像头与 IMU 之间的外参。数据序列文件夹包含 9 个序列的数据。

每个序列提供传感器数据、真实值和灰度图像。事件、图像和IMU数据以rosvbag格式提供, 包括两种类型的rosvbag: 一种图像为30Hz, IMU为1000Hz, 另一种图像为20Hz, IMU为200Hz。用户可以根据自己的算法选择合适的rosvbag, 并在需要时进行下采样操作。基线轨迹提供在CSV文件中。此外, 我们还发布了一个TUM格式的真实值文件, 以便使用evo工具进行评估。此外, 左右相机在30Hz帧率下的灰度图像已被提取并分别存储在l1和r1文件夹中, 同时附有时间戳对齐文件。数据序列组织如下:

```

id.sequence.name
├── data rosvbag
│   ├── name_imu_1000hz_images_30hz.bag
│   └── name_imu_200hz_images_20hz.bag
├── groundtruth
│   ├── gt.csv
│   └── gt.tum
└── Stereo images
    ├── l1
    ├── r1
    └── timestamp_pairs.txt

```

E. Baseline trajectory using visual SLAM

各种主流的SLAM系统在所提出数据集的九个序列上进行了测试, 详细特性如表II所示。为了评估性能, 我们采用绝对轨迹误差 (ATE) 作为主要指标。为了进行一致的比较, 我们使用evo工具来

将每个估计的轨迹与其对应的真实轨迹对齐, 从而计算精确的ATE测量值。量化结果如表III所示。

为了评估基于帧的视觉SLAM在水下环境中的性能, 使用Vins-Stereo-fusion [8] 和 ORB-SLAM2 [9] 进行测试。Vins的测试是使用带有200Hz IMU数据和20Hz图像的rosvbags进行的, 而ORB-SLAM2的测试使用30Hz的灰度图像。需要特别指出的是, 选择ORB-SLAM2进行测试的目的是评估视觉里程计 (VO) 与视觉惯性里程计 (VIO) 在所提出的水下数据集上的性能差异。最先进的事件相机SLAM算法ESVO2 [10] 也进行了测试, 以评估事件相机在具有挑战性的水下环境中的性能。

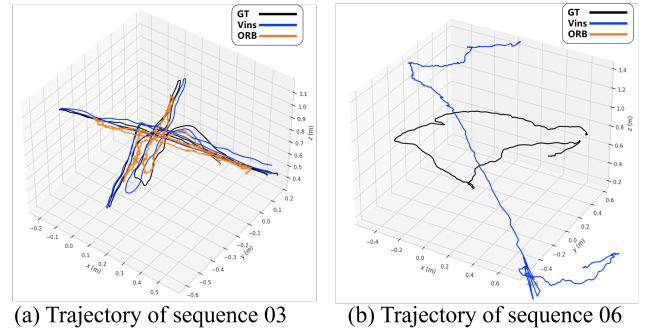


图2: 两个样本序列的估计轨迹和真实轨迹 (GT)。

1) Frame-based visual SLAM:

如表 III 所示, 基于简单类别序列 (S01-S05) 的测试结果表明, VIO 系统 (VINS-Stereo) 的姿态估计精度明显高于 VO 系统 (ORB-SLAM2)。这是因为水波和浮力的存在导致数据采集平台在转弯时出现高频快速抖动。一方面, 这违反了恒定速度

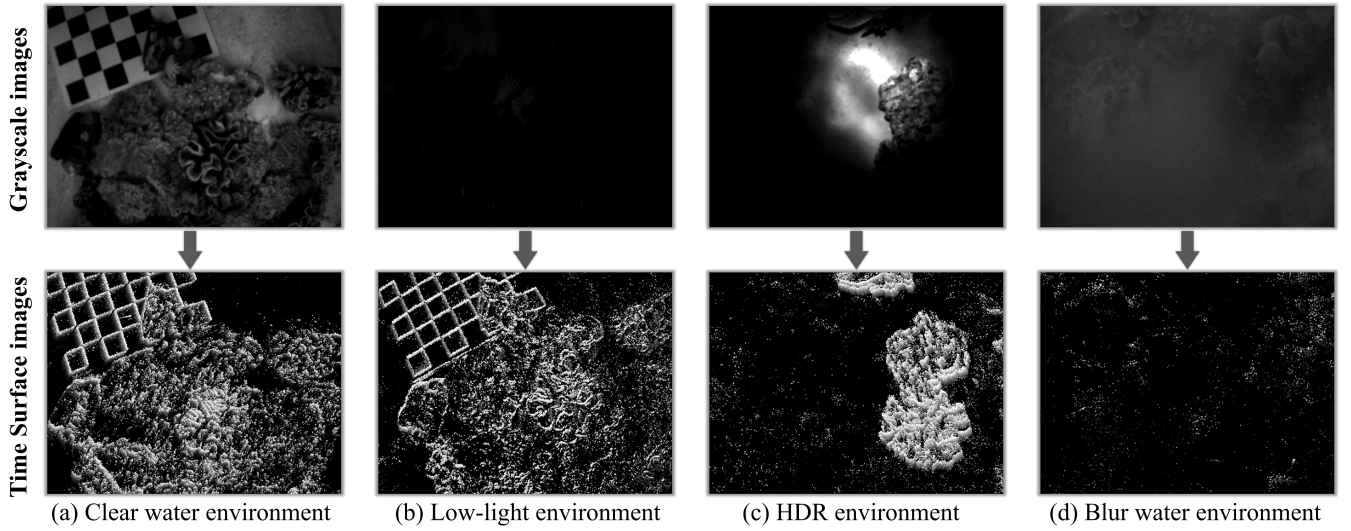


Fig. 3: Comparison of grayscale images and their corresponding Time Surface images under different underwater environmental conditions.

motion model assumption relied upon by ORB-SLAM2, making pose estimation no longer accurate. On the other hand, under the premise of similar features in underwater environment and lower camera resolution, small-range rapid movements can cause degradation in visual pose estimation. In fact, based on what was mentioned in SVin2 [11], this paper also applied CLAHE histogram equalization [12] to enhance visual features in the input images, but still did not achieve better results. VINS-Stereo, with the assistance of IMU data, can effectively compensate for the deficiencies in visual information. The IMU can provide high-frequency attitude and acceleration measurements, maintaining accurate pose estimation in the short term even when visual features are scarce. For data sequences with smoother motion and without rapid turns (S03), both ORB-SLAM2 and VINS can run relatively well, as shown in Fig. 2(a).

In experiments based on hard class sequences, both Vins-Stereo and ORB-SLAM2 exhibited different degrees of failure. Under low-light conditions (S06-S07), ORB-SLAM2 struggled to extract effective ORB feature points, making it impossible to estimate trajectories. VINS-Stereo, on the other hand, uses a combination of Shi-Tomasi corner detection with optical flow tracking for matching, which can detect more reliable corners in low-contrast images, as shown in Fig. 2(b). Additionally, optical flow tracking utilizes inter-frame continuity rather than re-detecting and matching features, reducing dependency on single-frame image quality, thus enabling more continuous trajectories. However, in HDR environment (S08), only a small area is illuminated, causing the scene to be in a state of constant abrupt changes, which can cause optical flow tracking to fail and affect VINS's performance. For blurry environment (S09), although ORB-SLAM2 had positioning errors of around 0.15m, it could not operate stably throughout the entire sequence, experiencing tracking loss during blurry frames. It was only through its

effective relocalization capability that it could obtain pose estimation results for clear frames.

To sum up, IMU data is essential for the application of visual SLAM system in underwater scenarios. Meanwhile, to ensure the performance of underwater visual odometry, it is also important to extract more robust visual feature points.

2) Event-based visual SLAM:

As shown in Table III, ESVO2 failed on all data sequences. The main reasons for ESVO2's failure can be attributed to: ESVO2 requires extremely high calibration precision, especially when distortion correction is not accurate enough, which significantly affects the system's precision. Although the Kalibr calibration tool used in this paper can effectively obtain intrinsic and extrinsic parameters on land, it does not include specific optimizations for underwater calibration. Underwater, light passes through three layers of refraction and reflection - water, transparent panel, and air - which to some extent affects the accuracy of camera calibration. Additionally, due to the low resolution of the DAVIS346 camera and our short stereo camera baseline, there were negative impacts on depth estimation results and overall accuracy. Finally, ESVO2 demands robust initialization with high-quality reconstruction, however, the cluttered underwater environment textures often lead to reconstruction failures, affecting system operation. It should be noted that this does not mean ESVO2 lacks good pose estimation capability, but only indicates that it cannot effectively address challenges in underwater scenes under this paper's experimental setup. In fact, ESVO2 has demonstrated excellent performance on land when using the long-baseline, high-resolution solution.

F. Event Imaging Under Adverse Conditions

Event cameras have unique advantages in visual representation of complex underwater scenes. Time Surface (TS) is a commonly used event representation method, with the following principle:

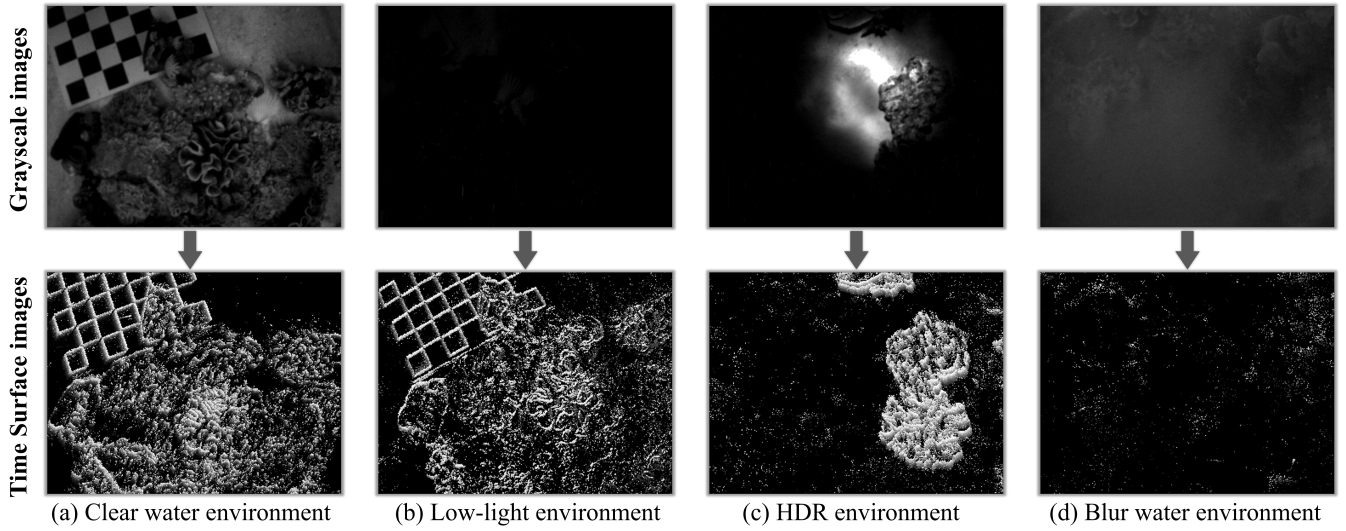


图3: 在不同水下环境条件下灰度图像及其对应时间表面图像的比较。

ORB-SLAM2所依赖的运动模型假设, 使得位姿估计不再准确。另一方面, 在水下环境中特征相似且相机分辨率较低的前提下, 小范围的快速运动会导致视觉位姿估计性能下降。事实上, 基于SVin2 [11]中提到的内容, 本文也应用了CLAHE直方图均衡化 [12] 来增强输入图像中的视觉特征, 但仍未取得更好的结果。VINS-Stereo在IMU数据的辅助下, 可以有效弥补视觉信息的不足。IMU可以提供高频率的姿态和加速度测量, 即使在视觉特征稀少的情况下, 也能在短期内保持准确的位姿估计。对于运动更加平稳且没有快速转向的数据序列 (S03), 如图2(a)所示, ORB-SLAM2和VINS均可以较好地运行。

在基于困难类别序列的实验中, VINS-Stereo 和 ORB-SLAM2 都表现出不同程度的失败。在低光照条件下 (S06-S07), ORB-SLAM2 难以提取有效的 ORB 特征点, 使得无法进行轨迹估计。而 VINS-Stereo 则结合了 Shi-Tomasi 角点检测和光流跟踪进行匹配, 这可以在低对比度图像中检测到更可靠的角点, 如图 2(b) 所示。此外, 光流跟踪利用帧间连续性而不是重新检测和匹配特征, 减少了对单帧图像质量的依赖, 从而实现了更连续的轨迹。然而, 在 HDR 环境 (S08) 下, 仅有小面积受到照明, 导致场景处于持续的剧烈变化状态, 这可能导致光流跟踪失败, 并影响 VINS 的性能。对于模糊环境 (S09), 虽然 ORB-SLAM2 的定位误差约为 0.15m, 但它无法

有效的重定位能力, 使其能够对清晰帧获得位姿估计结果。

总之, IMU 数据对于水下场景中视觉 SLAM 系统的应用至关重要。同时, 为了确保水下视觉里程计的性能, 提取更稳健的视觉特征点也很重要。

2) Event-based visual SLAM:

如表 III 所示, ESVO2 在所有数据序列上均失败。ESVO2 失败的主要原因可以归结为: ESVO2 需要极高的标定精度, 尤其是在畸变校正不够准确时, 这会显著影响系统的精度。虽然本文使用的 Kalibr 标定工具可以在陆地上有效获得内参和外参, 但其并未针对水下标定进行特定优化。在水下, 光线穿过水、透明板和空气三层折射和反射, 这在一定程度上影响了相机标定的精度。此外, 由于 DAVIS346 相机的分辨率较低以及我们的双目相机基线较短, 对深度估计结果和整体精度也产生了负面影响。最后, ESVO2 对高质量重建和稳健初始化有较高要求, 但在复杂水下环境中, 纹理的杂乱常常导致重建问题。

F. Event Imaging Under Adverse Conditions

事件相机在复杂水下场景的视觉表现上具有独特优势。时间表面 (TS) 是一种常用的事件表示方法, 其原理如下:

$$\mathcal{T}(\mathbf{x}, t) \doteq \exp\left(-\frac{t - t_{\text{last}}(\mathbf{x})}{\eta}\right), \quad (1)$$

where t_{last} is the timestamp of the last event at each pixel coordinate $\mathbf{x} = (u, v)^\top$, and η is a parameter controlling the decay rate. Time Surface uses an exponential decay kernel to emphasize recent events while attenuating past events.

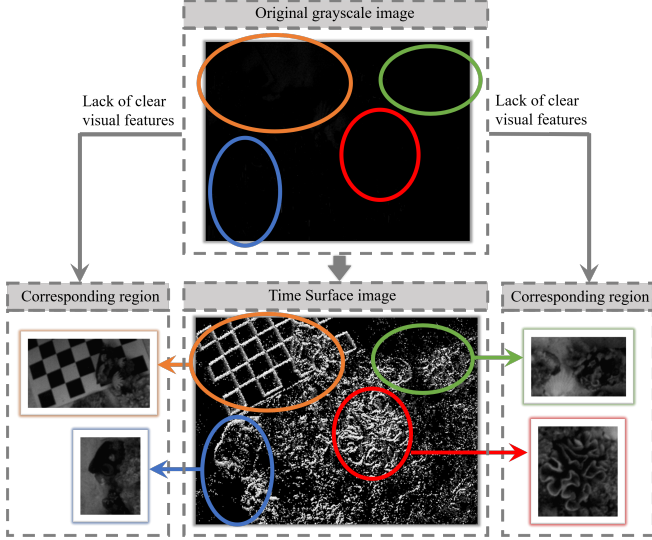


Fig. 4: Details comparison of Time Surface image and original grayscale imaging features in low-light underwater environment (Sequence 06).

The dataset published in this paper includes four scenarios: clear water, low light, HDR, and turbid water. Fig. 3 (a) shows the clear water condition, where the corresponding Time Surface effectively reflects the environmental texture features. For coral regions with low contrast, Time Surface can also represent their surface textures and contours. A detailed comparison of Time Surface images and original grayscale imaging features in low-light underwater environments is shown in Fig. 4. Fig. 3 (b) and (c) represent challenging lighting conditions. Evidently, Time Surfaces generated from event data demonstrate strong robustness to lighting conditions, effectively presenting environmental textures in low-light or HDR scenes.

Failure case: As shown in Fig. 3(d), Time Surfaces fail when water becomes turbid. This is because numerous impurities obscure environmental details, preventing event cameras from properly capturing changes in illumination. This phenomenon indicates that event cameras do not necessarily perform better than conventional cameras in certain scenarios.

Going beyond TS image: While Time Surface imaging effectively reveals features in clear and low-light conditions, more robust event representations are needed for challenging underwater environments. To be more specific, Motion-compensated Event Frame method [13] [14] that warps event data to a reference frame using motion models can significantly reduce the influence of suspended particles by aligning

events from real objects while dispersing random noise. Event Packet representations [15] preserves temporal precision by aggregating events within time windows, enabling batch operations that better distinguish between consistent features and sporadic noise in turbid waters. Voxel Grid [16] approach encodes events into 3D space-time tensor representations, allowing for structured spatial and temporal integration that enhances signal-to-noise ratio in environments with varying visibility. By combining the advantages of these methods, adaptive systems can be developed for different underwater conditions, enabling more reliable perception and localization in challenging underwater environments.

III. CONCLUSION

This paper presents an underwater dataset containing event, frame, and IMU data, along with groundtruth trajectories for evaluation purpose. The dataset is categorized into "easy" and "hard" classes, where sequences in the easy class were collected under good water quality and lighting conditions, while sequences in the hard class include special underwater conditions such as low light, high dynamic range (HDR), and turbidity. Furthermore, based on this dataset, we evaluated mainstream frame-based visual SLAM algorithms and a state-of-the-art event-based SLAM algorithm in underwater scenarios. The analysis of advantages and limitations of event camera in underwater environments points to potential future directions for underwater SLAM research. In the future, we will continue to improve and update this dataset, providing more realistic underwater scenes and more accurate calibration data, striving to contribute a comprehensive benchmark for underwater robotics research.

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$$\mathcal{T}(\mathbf{x}, t) \doteq \exp\left(-\frac{t - t_{\text{last}}(\mathbf{x})}{\eta}\right), \quad (1)$$

其中 t_{last} 是每个像素坐标 $\mathbf{x} = (u, v)^T$ 上的最后事件的时间戳, η 是控制衰减速率的参数。时间表面使用指数衰减核来强调最近的事件, 同时抑制过去的事件。

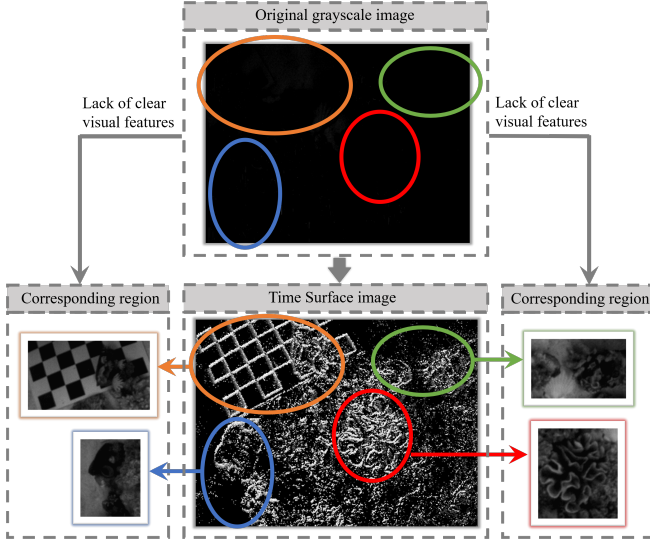


图4: 低光水下环境中 (序列06) 时间表面图像与原始灰度成像特征的细节比较。

本文发布的数据集包含四种场景: 清水、低光、HDR 和混浊水域。图 3(a) 显示了清水条件下的情况, 其中对应的时间表面有效地反映了环境纹理特征。对于低对比度的珊瑚区域, 时间表面也可以表现其表面纹理和轮廓。图 4 展示了低光水下环境中时间表面图像与原始灰度成像特征的详细对比。图 3(b) 和 (c) 表示具有挑战性的光照条件。显然, 由事件数据生成的时间表面对光照条件表现出较强的鲁棒性, 在低光或 HDR 场景中有效呈现环境纹理。

失败情况: 如图3(d)所示, 当水变得浑浊时, 时间表面会失效。这是因为大量杂质遮挡了环境细节, 阻止事件相机正确捕捉光照变化。这一现象表明, 在某些场景中, 事件相机不一定比传统相机表现更好。

超越 TS 图像: 虽然时间表面成像能够有效地在清晰和弱光条件下揭示特征, 但在具有挑战性的水下环境中, 需要更强大的事件表示。更具体地说, 运动补偿事件帧方法 [13][14] 通过使用运动模型将事件数据变换到参考帧, 可以通过对齐显著减少悬浮颗粒的影响

从真实物体产生事件, 同时分散随机噪声。事件包表示 [15] 通过在时间窗口内聚合事件来保持时间精度, 使批量操作能够更好地区分浑浊水域中一致的特征和零散的噪声。体素网格 [16] 方法将事件编码为三维时空张量表示, 允许进行结构化的空间和时间整合, 从而在能见度变化的环境中提高信噪比。通过结合这些方法的优势, 可以为不同的水下条件开发自适应系统, 从而在复杂的水下环境中实现更可靠的感知和定位。

三、结论

本文提出了一个水下数据集, 包含事件、帧和IMU数据, 以及用于评估的真实轨迹。该数据集分为“简单”和“困难”两类, 其中简单类的序列是在良好的水质和光照条件下采集的, 而困难类的序列包括低光、高动态范围 (HDR) 和浑浊等特殊水下条件。此外, 基于该数据集, 我们评估了主流帧式视觉SLAM算法和最先进的事件式SLAM算法在水下场景中的表现。对事件相机在水下环境中的优势和局限性的分析指出了水下SLAM研究的潜在未来方向。未来, 我们将继续改进和更新该数据集, 提供更加逼真的水下场景和更精确的标定数据, 努力为水下机器人研究提供一个全面的基准。

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